

COMPARATIVE ANALYSIS

1. Object Detection Performance Comparison

Two object detection models, Viola–Jones and YOLO, produce the following results: Model A has 80 True Positives, 20 False Positives, and 30 False Negatives, while Model B has 90 True Positives, 15 False Positives, and 20 False Negatives.

(a) Calculate the precision and recall for both models.

(b) Compare the results and determine which model provides better overall object detection performance. Justify your answer based on the calculated values.

2. Motion Estimation Error Analysis

Two motion estimation techniques produce the following displacement errors in pixels: Method A: 2, 3, 2 and Method B: 3, 5, 4 for three consecutive frames.

(a) Calculate the average displacement error for both methods.

(b) Compare the two methods and determine which method provides more accurate motion estimation.

(c) Justify your answer based on the error values.

3. Depth Estimation Comparison

A stereo vision system and a Structure-from-Motion (SfM) system estimate the depth of three objects. The estimated depths are: Stereo Vision: 2.0 m, 3.0 m, 4.0 m and SfM: 2.5 m, 3.8 m, 5.0 m, while the actual depths are 2 m, 3 m, and 4 m, respectively.

(a) Calculate the absolute error for each object for both methods. (b) Calculate the average absolute error for both methods.

(c) Compare the methods and identify which provides more accurate depth estimation.

1. Object Detection Performance Comparison

Given:

Model	TruePositives (TP)	False Positives (FP)	False Negatives (FN)
Model A (Viola–Jones)	80	20	30
Model B (YOLO)	90	15	20

(a) Calculate Precision and Recall

Formula for Precision

$$\text{Precision} = \frac{TP}{TP + FP}$$

Precision tells us how many of the detected objects are actually correct.

Formula for Recall

$$\text{Recall} = \frac{TP}{TP + FN}$$

Recall tells us how many of the actual objects were successfully detected.

Model A Precision

$$\text{Precision A} = \frac{80}{20 + 80}$$

$$=80/100$$

$$\text{Precision A} = 0.80 = 80\%$$

Recall

$$\text{Recall A} = 80/80+30$$

$$=80/110$$

$$\text{Recall A} \approx 0.7273 = 72.73\%$$

Model B

$$\text{Precision B} = 90/90+15$$

$$=90/105$$

$$\text{Precision B} = 0.8571 = 85.71\%$$

Recall B

$$\text{Recall B} = 90/90+20$$

$$=90/110$$

$$\text{Recall B} = 0.8182 = 81.82\%$$

Results

Model	Precision	Recall
Model A	80%	72.73%
Model B	85.71%	81.82%

(b) Comparison

Model B has:

Higher precision: $85.71\% > 80\%$

Higher recall: $81.82\% > 72.73\%$

More true positives: $90 > 80$

Fewer false positives: $15 < 20$

Fewer false negatives: $20 < 30$

Therefore,

Model B (YOLO) provides better overall object detection performance.

Justification: Model B detects more actual objects while producing fewer incorrect detections. It therefore performs better in both precision and recall.

2.Motion Estimation Error Analysis

Given:

Method A errors = 2, 3, 2 pixels

Method B errors = 3, 5, 4 pixels

(a) Calculate Average Displacement Error

Formula

Average Displacement Error= Sum of all displacement errors/Number of frames

Method A

Average A = $2+3+2/3$

Average A = $7/3 = 2.33$ pixels

Method B

Average B = $3+5+4/3$

Average B = 4 pixels

Results

Method	Errors (pixels)	Average Error
Method A	2, 3, 2	2.33 pixels
Method B	3, 5, 4	4 pixels

(b) Which method is more accurate?

Since displacement error should be as small as possible:

$$2.33 < 4$$

Therefore,

Method A is more accurate.

(c) Justification

Method A has an average displacement error of 2.33 pixels, while Method B has an average error of 4 pixels.

A lower displacement error means the estimated motion is closer to the actual motion.

Therefore,

Method A provides more accurate motion estimation.

3. Depth Estimation Comparison

Given:

Actual depths:

2m, 3m, 4m

Stereo Vision estimates: 2.0m, 3.0m, 4.0m

SfM estimates:

2.5m, 3.8m, 5.0m

(a) Calculate Absolute Error

Formula

$$\text{Absolute Error} = |\text{Estimated Depth} - \text{Actual Depth}|$$

The absolute value is used so that the error is always positive.

Stereo Vision

Object 1

Actual depth = 2 m

Estimated depth = 2.0 m

$$\text{Error} = |2.0 - 2|$$

$$\text{Error} = 0\text{m}$$

Object 2

Actual depth = 3 m

Estimated depth = 3.0 m

$$\text{Error} = |3.0 - 3|$$

$$\text{Error} = 0\text{m}$$

Object 3

Actual depth = 4 m

Estimated depth = 4.0 m

$$\text{Error} = |4.0 - 4|$$

$$\text{Error} = 0\text{m}$$

Therefore, Stereo Vision errors are:

0m, 0m, 0m

SfM

Object 1

Actual depth = 2 m

Estimated depth = 2.5 m

$$\text{Error} = |2.5 - 2|$$

$$\text{Error} = 0.5\text{m}$$

Object 2

Actual depth = 3 m

Estimated depth = 3.8 m

Error=|3.8-3|

Error=0.8m

Object 3

Actual depth = 4 m

Estimated depth = 5.0 m

Error=|5.0-4|

Error=1.0m

Therefore, SfM errors are:

0.5m, 0.8m, 1.0m

(b) Calculate Average Absolute Error

Formula

Average Absolute Error= $\sum$ Absolute Errors/Number of Objects

Stereo Vision

Average Error=0+0+0/3

Average error =0

SfM

Average Error = 0.5+0.8+1.0/3 =2.3/3

Average Error≈0.767m

(c) Comparison

Object	Actual Depth	Stereo Estimate	Stereo Error	SfM Estimate	SfM Error
1	2 m	2.0 m	0 m	2.5 m	0.5 m
2	3 m	3.0 m	0 m	3.8 m	0.8 m
3	4 m	4.0 m	0 m	5.0 m	1.0 m
Average	—	—	0 m	—	0.767 m

Since lower depth error indicates better accuracy:

$$0 < 0.767$$

Therefore,

Stereo Vision provides more accurate depth estimation.

Justification

Stereo Vision has an average absolute error of 0 m, meaning its estimated depths exactly match the actual depths for all three objects. SfM has an average absolute error of approximately 0.767 m.